

DEVANSHU CHICHOLIKAR

AI ENGINEER | MCP & AGENT INFRASTRUCTURE | BOSTON, MA (OPEN TO RELOCATION)

(857) 339-8803 | chicholika.d@northeastern.edu | linkedin.com/in/devanshuchicholika | github.com/DevanshuNEU | devanshuchicholika.com

AI / AGENT ENGINEERING PROJECTS

- OpenCodeIntel** | *Python, FastAPI, FastMCP, tree-sitter, Pinecone, Redis, Supabase, React, TS* **Sep 2024 - Present**
- Shipped a **production MCP server** exposing 12 tools (semantic code search, dependency graph, impact analysis, **context assembly**) over dual transport, stdio for local agents and streamable-HTTP for hosted **Claude.ai** connectors with Bearer-token middleware, letting AI agents (**Claude, Cursor**) retrieve codebase context at **p95 208ms**
 - Designed a **hybrid AST + embedding RAG** pipeline (tree-sitter function-level chunking, BM25 + vector fusion, cross-encoder **reranking** with graceful degradation); benchmarked to **94% average Hit@1** across **14 OSS codebases** (700-query eval), improving accuracy **+8.4 points** from reranking isolated via a **98-run ablation**
 - Hardened the public **REST API**: dual-mode auth (HS256 JWT + SHA-256 keys), two-layer **multi-tenant isolation** (Pinecone metadata filters + Postgres row-level security), Redis tiered rate limiting, and real-time indexing streamed over Redis pub/sub into authenticated **WebSocket** endpoints
 - Drove retrieval quality with **eval discipline**: preflight data-quality checks (+12% Hit@1), documented negative results, per-repo human-query sets; **700+ commits, 450+ tests, 159 merged PRs** solo over a 47K-LOC codebase with path-filtered CI (Trivy + TruffleHog scans)
- CallBudget** | *Python, FastMCP, scikit-learn, Optuna, Pipecat, Deepgram, DuckDB* **2026**
- Deployed an agentic system end-to-end** against a real pharmacy-stock workflow: a **FastMCP server** (predict / plan / eval / converse) over a learned stock-probability ranker, cutting expected calls-to-find from **4.3 to 2.3 (47%)** on a 19-pharmacy simulation
 - Designed a **calibrated-abstention** voice layer (self-consistency voting) that drove false-positive in-stock answers from **10% to 0%** under a deliberately unreliable extractor, the **guardrail** layer that makes an agent safe in front of users
 - Hardened the tuning loop with **Optuna** seeded-TPE under a hard safety constraint and a held-out validation guard that rejected its own overfit gains; one engine serves **MCP, CLI, and a FastAPI web UI** with a Claude-driven live voice agent
- tool-crowding** | *Author | Python, MCP, Anthropic API, pytest* | MCP agent-evaluation research (Apache 2.0) **2026**
- Authored a **pre-registered open-methodology benchmark** for MCP tool-selection interference: a **345-test fail-closed harness**, 199-tool synthetic corpus, and 144-trial factorial design locked before data collection for fully reproducible, cache-clean measurement
 - Engineered **byte-level reproducibility** for LLM experiments: per-trial nonces defeat prompt caching, a runtime assertion halts any run on a cache hit, and server SHAs + tool schemas hash into every run ID
 - Falsified** the naive more-tools-degrade-routing hypothesis with exploratory probes (0 mis-routing at N=6 dissimilar tools), isolating the real failure mode to a task-ambiguity by agent-persona interaction
- Saar** (Chrome extension) | *TypeScript, WXT, Chrome MV3, React, Vitest, Playwright* **Mar 2026 - Present**
- Shipped a **Chrome MV3 extension** to the Chrome Web Store that tracks Claude.ai token usage and cost in real time, **entirely client-side** (no backend, nothing leaves the browser)
 - Built a three-context architecture, a MAIN-world fetch interceptor decoding **Anthropic SSE streams**, a Shadow-DOM overlay over validated postMessage, and a service worker running a BPE tokenizer (js-tiktoken), reading exact input_tokens for measured **context-window** tracking; **1,808 Vitest tests** across 63 files plus Playwright e2e
- saar CLI** | *Creator | Python* | published on PyPI (22 releases) **Apr 2026**
- Published a **Python CLI to PyPI** that statically analyzes a codebase (package manager, logging, auth patterns) and generates **agent context files** (AGENTS.md, CLAUDE.md, .cursorsrules)
- SecureScale** | *AWS, Terraform, Packer, Docker, Lambda, RDS, KMS, CloudWatch* **2025**
- Provisioned **multi-AZ AWS infrastructure** (VPC, ALB, ASG, RDS, S3 with KMS) as modular **Terraform** with **IAM least-privilege** defense-in-depth, cutting provisioning from **2 hours to 10 minutes** and **cloud spend 30% at 99.9% uptime** with CloudWatch

EXPERIENCE

- Parsewave** | **frontier-model data vendor** **May 2026 - Present**
Terminal-Bench Task Author (Contract) **Remote**
- Authored **Harbor-validated agent-evaluation environments** under contract (model training/eval data): 4 agent-debugging tasks with oracle/no-op verification and **anti-cheat test design**
 - Designed a reverse-engineering task that held **Claude Opus 4.8 to 0/5 resolved** across 5 graded trials, passing AI review to human review
- Northeastern University** **Sep 2025 - May 2026**
Graduate Teaching Assistant, Cloud Computing & Networks **Boston, MA**
- Taught **AWS, Terraform, and distributed systems** to **60+ graduate students**; authored a **Docker + GitHub Actions CI/CD lab** adopted as official course content across 3 sections (**180+ students**)
 - Led **system-design and code reviews** for **15 cloud-native API project teams**, coaching on scalability, fault tolerance, and API design
- Jaksh Enterprise** **Aug 2022 - Jul 2024**
Software Engineer | *B2B industrial-equipment platform (contract)* **Ahmedabad, India**

- Developed a **Java / Spring Boot** product-catalog and quotation engine for **590+** customizable products serving a **10K+** **B2B customer base**, over PostgreSQL and a rules-based pricing module
- Cut quote-page **p95 latency 65%** (800ms to 280ms) via PostgreSQL indexing, Redis caching, and async processing
- Owned delivery end-to-end as one of **two contract engineers**: gathered client requirements, scoped functional specs, and shipped via Dockerized **CI/CD**, cutting release cycles from **2 weeks to 3 days**

Pitney Bowes

Jan 2022 - Jul 2022

Software Development Engineer Intern

Pune, India

- Built **REST APIs** for PitneyShipPro with idempotent endpoints, eliminating **40%** of manual data entry; raised automated **test coverage to 85%** with Jest and Cypress

SKILLS

AI-Assisted Development: **Claude Code** (subagents, hooks, custom skills, slash commands), **MCP** (Model Context Protocol) server development, **agentic workflows**, spec-first development with AI agents, LLM evals; Cursor

AI / Agents / Eval: MCP servers (FastMCP), agent tool-calling, **RAG** (hybrid AST + embedding, BM25 + reranking), **LLM evaluation** (pre-registration, calibrated abstention, self-consistency, guardrails, LLM-as-judge), voice agents (Pipecat, Deepgram), tree-sitter, **context engineering**

Languages: Python, TypeScript, Java, JavaScript, Go, SQL

Backend / APIs: FastAPI, Spring Boot, Node.js, REST, gRPC, WebSocket, microservices, distributed systems

Cloud / DevOps: AWS, GCP (Cloud Run, Pub/Sub), Docker, Kubernetes, Terraform, GitHub Actions, CI/CD, Railway, Vercel; observability (Sentry, CloudWatch)

Data: PostgreSQL, Redis, Pinecone, DuckDB, Supabase, Firestore, MongoDB

Frontend: React, Next.js, Tailwind CSS

Foundations: data structures & algorithms, distributed systems, OOP & design patterns, system design

OPEN SOURCE & AI ENGINEERING PRACTICE

Run a **spec-first AI-agent development workflow** daily: technical specifications and ADRs before code, **Claude Code** agents (custom subagents, hooks, skills) implement against them, human review before merge; **MCP server development across 3 projects**.

2,203 GitHub contributions in the last 12 months (1,660 commits, 217 PRs, 169 issues) across MCP servers, agent evals, and developer tooling; accepted bug report on **sageox/ox** (Go); docs contribution to **entireio/external-agents** under review.

EDUCATION

Northeastern University | GPA: 3.85

Sep 2024 - May 2026

M.S. in Software Engineering Systems

Boston, MA

Coursework: *Generative AI, MLOps, Distributed Systems, Cloud Computing, Database Management Design, Algorithms, Object-Oriented Design & Design Patterns*

Dhirubhai Ambani Institute of ICT

Aug 2018 - May 2022

B.Tech in Information and Communication Technology

Gujarat, India